

Joseph Nounagnon

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OBJECTIVE

Data Science major at UNC Charlotte graduating December 2027, seeking a Summer 2027 internship in technology and analytics. Build analytical pipelines end to end, from API collection through cleaning, validation, statistical modeling, and diagnostics, and communicate technical results in clear, accessible language.

EDUCATION

University of North Carolina at Charlotte

August 2025 to December 2027

Bachelor of Science in Data Science

Central Piedmont Community College

Graduated May 2025

Associate of Science | President's List | Phi Theta Kappa Honor Society

SKILLS

Languages: Python, SQL, Java

Libraries & Tools: Pandas, NumPy, statsmodels, scikit-learn, Matplotlib, Seaborn, nflreadpy, nba_api, Jupyter Notebooks, Git/GitHub, Microsoft Excel, Microsoft Office

Concepts: Exploratory Data Analysis, Statistical Analysis, Regression Modeling & Diagnostics (residual, Q-Q, Cook's distance, VIF, stepwise/AIC selection), Predictive Modeling, Feature Engineering, Train/Test Validation, Confidence Intervals, Data Cleaning & Validation, API Data Collection, Data Visualization, Technical Documentation

PROJECTS

Bank of America Stadium Economic Impact Analysis | Python | statsmodels | Pandas | Seaborn | UNCC Data and Society course

- Modeled 6,758 city years of municipal finance data with a five person team to test whether hosting an NFL franchise raises city spending per resident, informing a funding recommendation on the proposed \$650M public investment in Bank of America Stadium renovations.
- Built OLS regression models with year fixed effects, log transforming a right skewed outcome variable and dropping a collinear predictor flagged by correlation and VIF checks; the final model reached an adjusted R^2 of 0.789 with a significant NFL coefficient ($p < 0.001$).
- Validated the model with residual, Q-Q, and Cook's distance diagnostics, applied HC3 robust standard errors after detecting non constant variance, and confirmed the result held against a stepwise AIC selected model before recommending fully private funding.

Turf vs. Grass: NFL Lower-Extremity Injury Rate Analysis | Exploratory Data Analysis | Python | Pandas | nflreadpy | Matplotlib | UNCC Modeling and Society course

- Caught and corrected a denominator error that had reversed my own conclusion: injury report length grew from 8.0 to 11.4 listed players across a season while lower extremity entries held flat near 6, and temperature correlated with week number at -0.649, manufacturing a temperature effect that was really the season calendar.
- Built a 2,256 team week panel from five NFL seasons (2021 to 2025) of nflreadpy schedule and injury report data, joining each game to the next week the team actually played so bye weeks and rows dropped for missing temperature could not be misread as injury free weeks.
- Reported a null result with 95% confidence intervals rather than a marginal one: turf averaged 6.00 lower extremity entries per team week against 6.13 on grass (difference -0.12, CI [-0.35, +0.11]) with the sign flipping across five temperature bins, and documented the surface by roof confound (648 of 804 indoor team weeks were turf) as a limit on the claim.

NBA Fantasy Point Forecasting Model | Python | scikit-learn | statsmodels | nba_api | Independent Project

- Built a forecasting pipeline end to end in Python: pulled two NBA seasons through the nba_api with rate limiting, error handling, and local caching, applied a custom league scoring function verified by hand against real game logs, and merged the output into 243 paired player seasons.
- Caught and removed All Star Game contamination by cross checking a date based filter against the API's own season classification, and validated the 20 game, 15 minute inclusion threshold through a standard error analysis rather than accepting it by convention.
- Tested four model versions against a fixed baseline on a held out set and reported the negative finding that seven advanced rate stats overfit and lowered test R^2 from 0.766 to 0.743, selecting the simpler prior average plus age model (test R^2 0.766, MAE 5.77).

EXPERIENCE

Customer Service Associate | Lowe's Home Improvement, Matthews, NC

October 2024 to Present

Flooring, Paint & Lawn/Garden Departments

- Interpret technical product specifications and tradeoffs to guide customers through high cost purchasing decisions, translating complex detail into clear recommendations for non technical audiences.
- Track and manage inventory across three departments using in store stock systems, monitoring product availability and identifying replenishment needs to maintain department standards.

VOLUNTEER & COMMUNITY

Missionaries of the Poor, International Service Trip

August 2024

- Traveled to Jamaica for a week long structured service program, leading daily meal distribution, hygiene assistance, and recreational activities for 100 to 200 homeless, mentally ill, and severely impoverished individuals.

Youth Group Volunteer | St. Thomas Aquinas Catholic Church

Ongoing

- Help coordinate multiple community outreach events annually, organizing volunteers and supporting local service initiatives.